

ITA0609-Machine Learning

ASSESSMENT TOOL 1 – Application-Based Questions

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Question-1

A parking facility has three sections containing cars of different colours. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C.

Code:

```
# Bayes' Theorem - Parking Facility Problem

# Number of cars in each section
A = {"Red":5, "Blue":3, "Black":2}
B = {"Red":4, "Blue":6, "Black":5}
C = {"Red":7, "Blue":2, "Black":6}

# Probability of selecting each section
P_A = 1/4
P_B = 1/2
P_C = 1/4

# Probability of selecting a blue car from each section
P_Blue_A = A["Blue"] / sum(A.values())
P_Blue_B = B["Blue"] / sum(B.values())
P_Blue_C = C["Blue"] / sum(C.values())

# Total probability of selecting a blue car
P_Blue = (P_Blue_A * P_A) + (P_Blue_B * P_B) + (P_Blue_C * P_C)

# Bayes' Theorem
P_C_given_Blue = (P_Blue_C * P_C) / P_Blue

print("Probability of Blue from Section A:", round(P_Blue_A,4))
print("Probability of Blue from Section B:", round(P_Blue_B,4))
print("Probability of Blue from Section C:", round(P_Blue_C,4))
print()

print("Total Probability of Blue Car =", round(P_Blue,4))
print("Probability that Blue Car came from Section C =",
round(P_C_given_Blue,4))
print("Percentage =", round(P_C_given_Blue*100,2), "%")
```

Output:

```
Probability of Blue from Section A: 0.3
Probability of Blue from Section B: 0.4
Probability of Blue from Section C: 0.1333

Total Probability of Blue Car = 0.3083
Probability that Blue Car came from Section C = 0.1081
Percentage = 10.81 %
```

Question-2

Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

- (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
- 3.(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
- 5.(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
- 6.(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
- 7.(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
9. (i) Identify the epoch at which the model begins to overfit.
10. (ii) Suggest two effective techniques to reduce overfitting in this model.
11. (iii) Discuss how overfitting affects the model's generalization

Code:

```
import matplotlib.pyplot as plt

epochs = [1,2,3,4,5,6,7,8]

training_loss = [2.8,2.5,2.2,2.0,1.8,1.6,1.5,1.3]
validation_loss = [2.4,2.2,2.0,2.1,2.3,2.6,2.9,3.2]

plt.figure(figsize=(8,5))
plt.plot(epochs, training_loss, marker='o', label="Training Loss")
plt.plot(epochs, validation_loss, marker='o', label="Validation Loss")

plt.xlabel("Epoch")
plt.ylabel("Loss")
```

```

plt.title("Training vs Validation Loss")
plt.legend()
plt.grid(True)
plt.show()

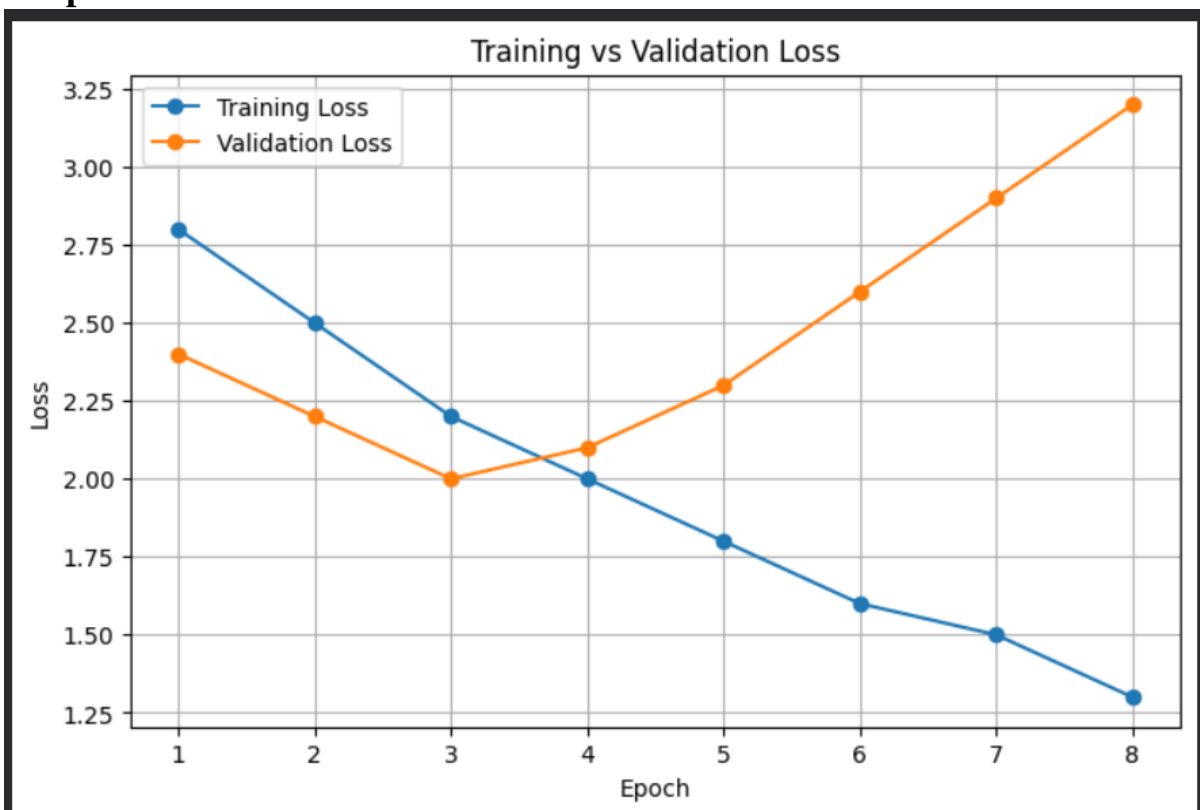
# Detect Overfitting
for i in range(1, len(validation_loss)):
    if validation_loss[i] > validation_loss[i-1]:
        print("Model starts overfitting at Epoch", epochs[i])
        break

print("\nTechniques to Reduce Overfitting:")
print("1. Early Stopping")
print("2. Dropout / L2 Regularization")

print("\nEffect:")
print("Overfitting decreases generalization performance on unseen data.")

```

Output:



Model starts overfitting at Epoch 4

Techniques to Reduce Overfitting:

1. Early Stopping
2. Dropout / L2 Regularization

Effect:

Overfitting decreases generalization performance on unseen data.

Question-3

Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: 12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58% 13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63% 14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68% 15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72% 16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73% 17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72% 18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70% 19. (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68% 20. i) Identify the epoch at which the model starts overfitting. 21. ii) Explain why validation accuracy decreases despite training accuracy increasing. 22. iii) Suggest two methods to improve validation performance.

Code:

```
import matplotlib.pyplot as plt

epochs = [1,2,3,4,5,6,7,8]

training_accuracy = [60,65,70,75,80,85,88,92]
validation_accuracy = [58,63,68,72,73,72,70,68]

plt.figure(figsize=(8,5))
plt.plot(epochs, training_accuracy, marker='o', label="Training Accuracy")
plt.plot(epochs, validation_accuracy, marker='o', label="Validation Accuracy")

plt.xlabel("Epoch")
plt.ylabel("Accuracy (%)")
plt.title("Training vs Validation Accuracy")
plt.legend()
plt.grid(True)
plt.show()

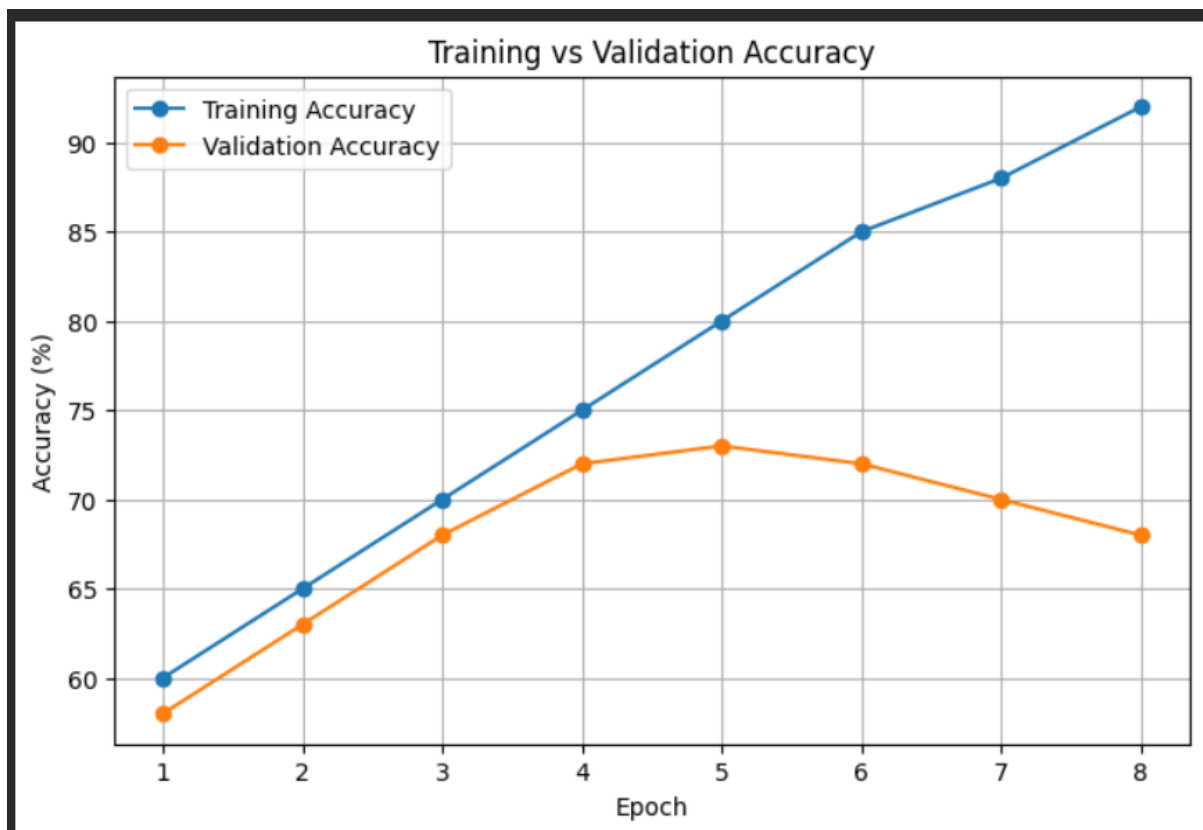
# Detect Overfitting
best = max(validation_accuracy)
best_epoch = validation_accuracy.index(best)

print("Best Validation Accuracy:", best,"%")
print("Overfitting starts at Epoch", epochs[best_epoch+1])

print("\nReason:")
print("Training accuracy increases while validation accuracy decreases because")
print("the model memorizes training data and fails to generalize.")

print("\nMethods to Improve Validation Performance:")
print("1. Early Stopping")
print("2. Data Augmentation / Regularization")
```

Output:



Best Validation Accuracy: 73 %

Overfitting starts at Epoch 6

Reason:

Training accuracy increases while validation accuracy decreases because the model memorizes training data and fails to generalize.

Methods to Improve Validation Performance:

1. Early Stopping
2. Data Augmentation / Regularization